REPORT:

1. Abstract

MindScope is an AI-based application designed to detect overthinking patterns in textual input. It uses Natural Language Processing (NLP) and machine learning techniques to analyze user-written thoughts and classify them as overthinking or normal thinking. The system also provides a confidence score and simple suggestions to improve mental clarity.

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2. Introduction

Overthinking is a common cognitive pattern where individuals repeatedly analyze thoughts, often leading to stress and anxiety. This project aims to build a simple and accessible AI tool that can identify such patterns in text and raise awareness.

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3. Objectives

- Detect overthinking from textual input
- Apply NLP techniques for text analysis
- Build a simple and interactive user interface
- Provide helpful feedback to users

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4. Methodology

4.1 Data Collection

A custom dataset was created with labeled sentences:

- 1 → Overthinking
- 0 → Normal thinking

4.2 Data Processing

- Text cleaned and prepared
- Converted into numerical format using TF-IDF

4.3 Model Used

- Logistic Regression
- Chosen for simplicity and efficiency

4.4 Working

1. User enters text
2. Text is vectorized
3. Model predicts class
4. Output + probability displayed

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5. Technologies Used

- Python
- Scikit-learn
- Pandas
- Streamlit
- Pickle

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6. System Architecture

Input Text → Preprocessing → TF-IDF Vectorization → ML Model → Prediction Output → User Interface

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7. Features

- Real-time text analysis
- Overthinking detection
- Probability score
- Simple mental wellness suggestions

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8. Results

The model successfully identifies patterns like:

- Repetitive thinking
- “What if” scenarios
- Negative loops

Accuracy is limited due to small dataset but works effectively for demonstration.

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9. Limitations

- Small dataset
- Basic model (not deep learning)
- Limited contextual understanding

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10. Future Enhancements

- Use advanced models (BERT)
- Expand dataset
- Add emotion tracking
- Integrate with chat platforms

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11. Conclusion

MindScope demonstrates how AI can be applied to mental awareness. It provides a simple yet effective way to detect overthinking patterns and encourages better thought management.

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12. References

- Scikit-learn Documentation
- NLP Basics
- Machine Learning Concepts